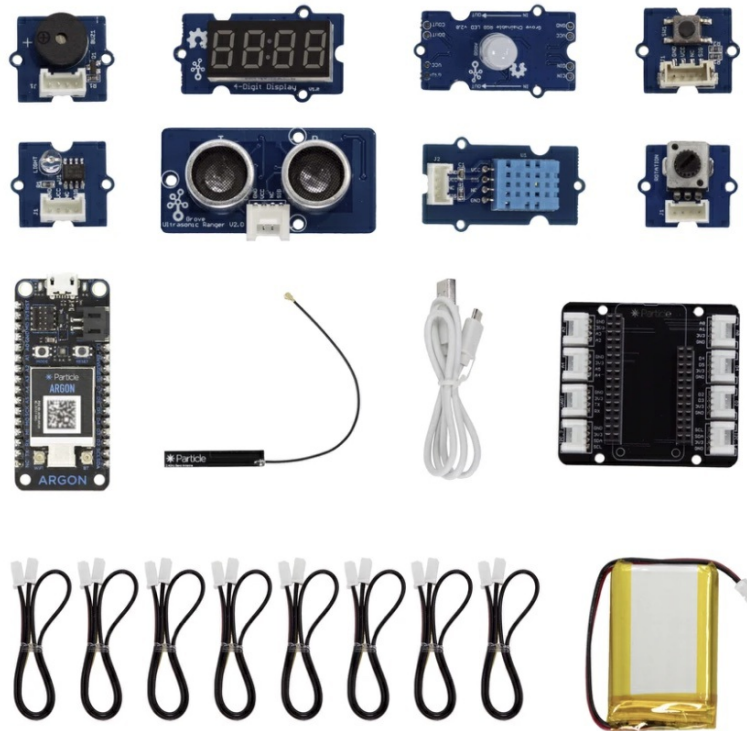


Gen 3 accessories	2
Grove starter kit	3
KIT CONTENTS	3
Grove shield	3
Button	4
Rotary angle sensor	4
Ultrasonic ranger	5
Temperature and humidity sensor	5
Light sensor v1.2	5
Chainable RGB LED	5
Buzzer	6
4-Digit Display	6
Adafruit FeatherWing OLED Display 128x32	7
Ethernet FeatherWing	8
PIN MAP	8
PoE (Power over Ethernet)	10
Classic adapter	12
Pin orientation	12
Substituting for an Electron	14
Substituting for an Photon	15
Schematic	16
Top	16
Bottom	17
PIN MAP	17
I2C	18
SPI	18
OTHER FEATURES	18
Debugger	20
FeatherWing Tripler	21

Gen 3 accessories

These accessories are for the Particle Gen 3 devices: Argon and Boron in the Adafruit Feather form-factor.

Grove starter kit



The Grove starter kit is designed to make your first exploration into the world of sensing and actuating a breeze. The kit comes with seven different components that work out-of-the-box with Particle Gen 3 hardware and let you focus on learning the basics. Working with Starter Kit requires no soldering and minimal wiring. Simply connect the kit's Grove Shield (the new co-developed FeatherWing adapter from Particle and SeeedStudio) to your Particle Gen 3 board and use the included cables to connect the adapter and the sensors and actuators that you want to use. It's a plug-and-play experience.

KIT CONTENTS

- 1x Grove Shield for Particle Gen 3 devices
- 1x Button
- 1x Rotary Angle Sensor
- 1x Ultrasonic Ranger
- 1x Temp&Humi Sensor
- 1x Light Sensor v1.2
- 1x Chainable RGB LED
- 1x Buzzer
- 1x 4-Digit Display

Grove shield

The main shield consists of a connector to plug in any of the Particle Argon or Boron dev kit and a set of 8 grove ports for plugging in various grove accessories. The ports consist of:

- 3 Digital ports
- 2 Analog ports
- 2 I2C ports
- 1 UART port

The pins on the Grove connector are as follows:

Pin	Color	Digital	Analog	I2C	UART
1	Yellow	Primary I/O	Primary Analog In	I2C Clock	Serial RX
2	White	Secondary I/O	Secondary Analog In	I2C Data	Serial TX
3	Red	Power	Power	Power	Power
4	Black	GND	GND	GND	GND

For a short (~6 min) example of using the Grove Starter Kit, watch the video below.

Button



Button is a momentary push button. It contains one independent "momentary on/off" button. "Momentary" means that the button rebounds on its own after it is released. The button outputs a HIGH signal when pressed, and LOW when released. The Sig marked on silk layer stands for signal while NC is not connected to anything.

It can be used on either a digital or analog input port, and the button is read using the primary digital I/O using the [digitalRead](#) function. The digitalRead function can be used on both digital and analog pins.

Additional information available [here](#).

Rotary angle sensor



The rotary angle sensor produces analog output between 0 and Vcc (3.3VDC) on its D1 connector. The D2 connector is not used. The angular range is 300 degrees with a linear change in value. The resistance value is 10k ohms. This essentially is a potentiometer.

It must be used on an analog input port, and the value is read using the primary analog input using the [analogRead](#) function.

Additional information available [here](#).

Ultrasonic ranger



Ultrasonic ranger is a non-contact distance measurement module which works at 40KHz. When we provide a pulse trigger signal with more than 10uS through signal pin, the sensor will issue 8 cycles of 40kHz cycle level and detect the echo. The pulse width of the echo signal is proportional to the measured distance. Here is the formula: Distance = echo signal high time * Sound speed (340M/S)/2. The sensor's trig and echo signal share 1 SIG pin.

The library is available in the Web IDE as [Grove_Ultrasonic_Ranger](#).

Additional information available [here](#).

Temperature and humidity sensor



This Temperature and Humidity sensor provides a pre-calibrated digital output. A unique capacitive sensor element measures relative humidity and the temperature is measured by a negative temperature coefficient (NTC) thermistor. It has excellent reliability and long term stability. Please note that this sensor will not work for temperatures below 0 degrees. It uses the DHT-11 sensor.

The library is available in the Web IDE as [Grove_Temperature_And_Humidity_Sensor](#).

Additional information available [here](#).

Light sensor v1.2



Light sensor integrates a photo-resistor (light dependent resistor) to detect the intensity of light. The resistance of photo-resistor decreases when the intensity of light increases. A dual OpAmp chip LM358 on board produces voltage corresponding to intensity of light(i.e. based on resistance value). The output signal is analog value, the brighter the light is, the larger the value.

It must be used on an analog input port, and the value is read using the primary analog input using the [analogRead](#) function.

This module can be used to build a light controlled switch i.e. switch off lights during day time and switch on lights during night time.

Additional information available [here](#).

Chainable RGB LED



Chainable RGB LED is based on P9813 chip which is a full-color LED driver. It provides 3 constant-current drivers as well as modulated output of 256 shades of gray. It communicates with a MCU using 2-wire transmission (Data and Clock). This 2-wire transmission can be used to cascade additional Grove - Chainable RGB LED modules. The built-in clock regeneration enhances the transmission distance. This Grove module is suitable for any colorful LED based projects.

The library is available in the Web IDE as [Grove_ChainableLED](#).

Additional information available [here](#).

Buzzer



Buzzer module has a piezo buzzer as the main component. The piezo can be connected to digital outputs, and will emit a tone when the output is HIGH. Alternatively, it can be connected to an pulse-width modulation output to generate various tones and effects.

You can use the [pinMode](#) and [digitalWrite](#) functions to control the output.

```
pinMode(D2, OUTPUT);  
digitalWrite(D2,  
HIGH);  
delay(1000);  
digitalWrite(D2, LOW);
```

It can be used on a digital or analog port, as the analog inputs can also be used as digital outputs.

An example using [tone](#) (PWM) to play a melody can be found [here](#). Not every pin is PWM compatible, so make sure you use a port that supports PWM.

Additional information available [here](#).

4-Digit Display

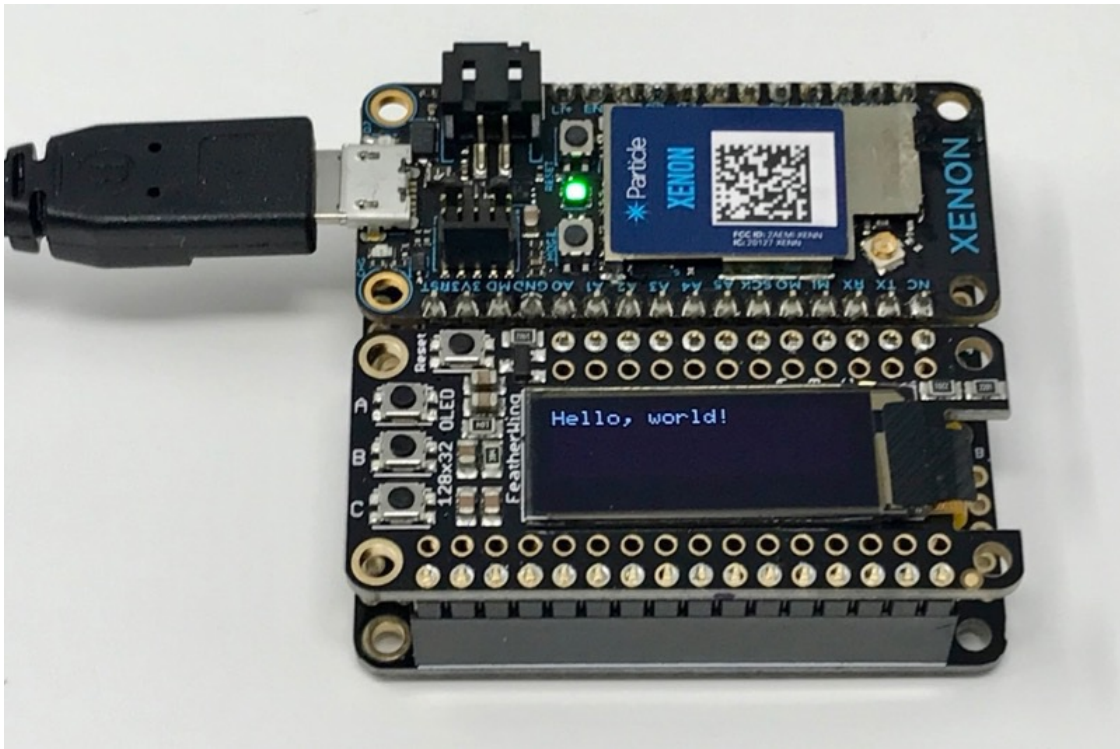


This is a 4 digit seven-segment display module that uses TM1637 driver chip. This driver is controlled with a two wire interface (CLK, DATA). This is perfect for displaying 4 digit numeric values.

The library is available in the Web IDE as [Grove_4Digit_Display](#).

Additional information available [here](#).

Adafruit FeatherWing OLED Display 128x32



The Adafruit OLED display FeatherWing makes it easy to add a small text and graphics display to your project. Built in a FeatherWing form-factor, you can use a FeatherWing Doubler or FeatherWing Tripler to add a display to your Argon or Boron project with no loose wires. And it includes three handy momentary push-buttons.

You can purchase one from [Adafruit](https://www.adafruit.com/product/4017). You can find more technical information [at Adafruit](https://www.adafruit.com/product/4017).

The library is available in the Web IDE as [oled-wing-adafruit](https://www.adafruit.com/product/4017). You can find additional documentation [here](https://www.adafruit.com/product/4017).

The display is a SSD1306 and connects by I2C (pins D0 and D1).

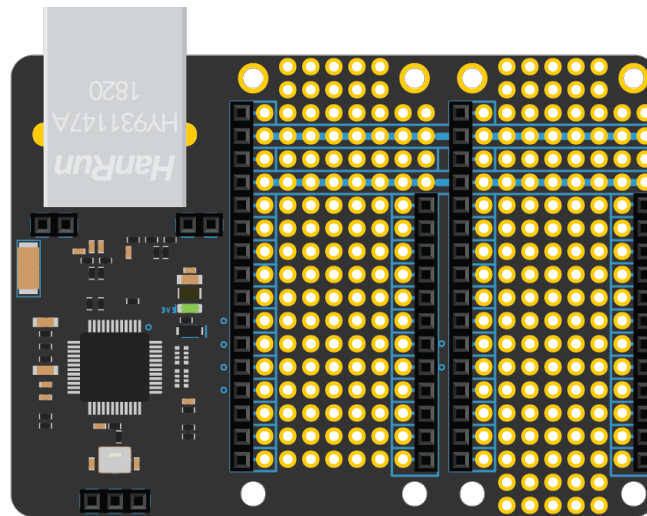
The three buttons are:

Button	Pin	Notes
A	D4	No pull-up. Can't be used with Ethernet.
B	D3	100K pull-up. Can't be used with Ethernet.
C	D2	No pull-up.

The library takes care of setting the appropriate input modes and debouncing the buttons for you.

Note that if you are using the Adafruit OLED display and the Particle Ethernet FeatherWing, you cannot use buttons A or B as those pins are used by Ethernet.

Ethernet FeatherWing

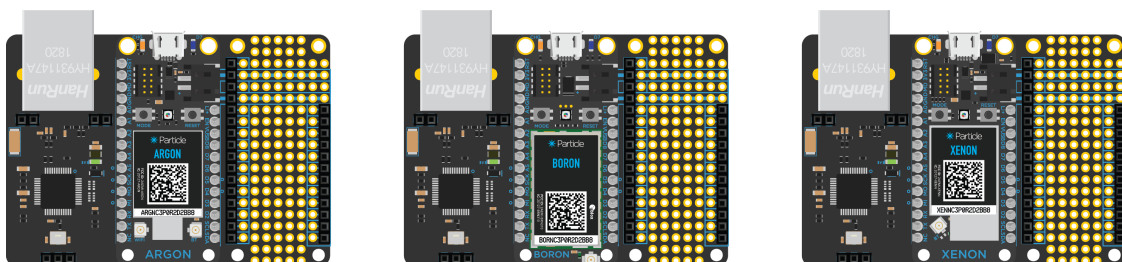


The Ethernet Featherwing has been deprecated and is no longer available for sale. The [Adafruit Ethernet Featherwing](#) can be used instead by following the instructions [here](#).

The Ethernet FeatherWing is the fastest way to add wired connectivity to your Argon or Boron and turns it into an Ethernet gateway. Based on the [WIZnet W5500](#) chip, this side-by-side FeatherWing maintains great RF performance even while adding Ethernet connectivity to any Particle Gen 3 device.

The form-factor is based around the Adafruit FeatherWing Tripler. The two side connectors allow you to plug in one Particle Gen 3 device along with a Feather accessory.

- 10BaseT/100BaseTX Ethernet
- Support Auto Negotiation (Full and half duplex, for both 10BaseT and 100BaseTX)



PIN MAP

Particle Pin	Ethernet FeatherWing Pin
MISO	SPI MISO
MOSI	SPI MOSI
SCK	SPI SCK
D3	nRESET ^[1]
D4	nINTERRUPT ^[1]

Notes:

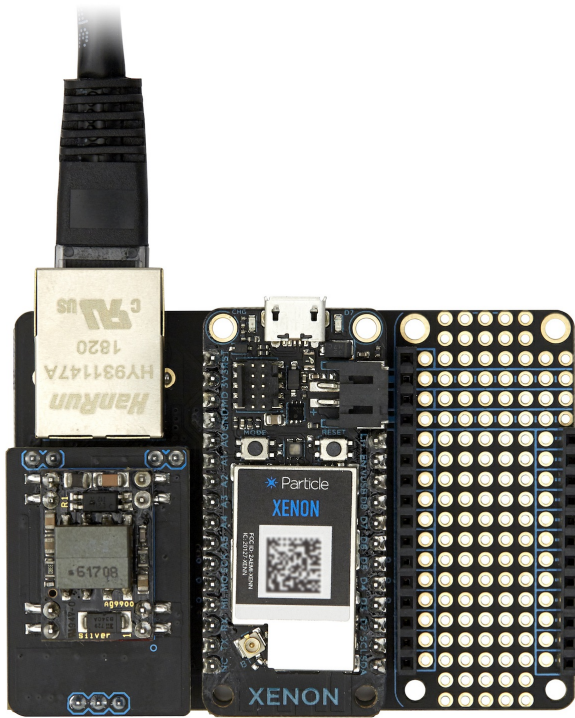
^[1] These pins are connected via jumper pads on the bottom. You can cut these jumper traces and rewire them to a different pin on the Gen 3 device if the need arises. However the Device OS does not allow use of other pins (at least through 1.1.0).

The hardware design for this wing is completely open source. All the files are available on our [GitHub repository here](#).

If you are using the Adafruit Ethernet Feather Wing, be sure to connect the nRESET and nINTERRUPT pins (on the small header on the short side) to pins D3 and D4 with jumper wires. These are required for proper operation.

For more information about Ethernet, see the application note [AN037 Ethernet](#).

PoE (Power over Ethernet)



The PoE adapter is discontinued and no longer available for sale. Existing PoE adapters will continue to work.

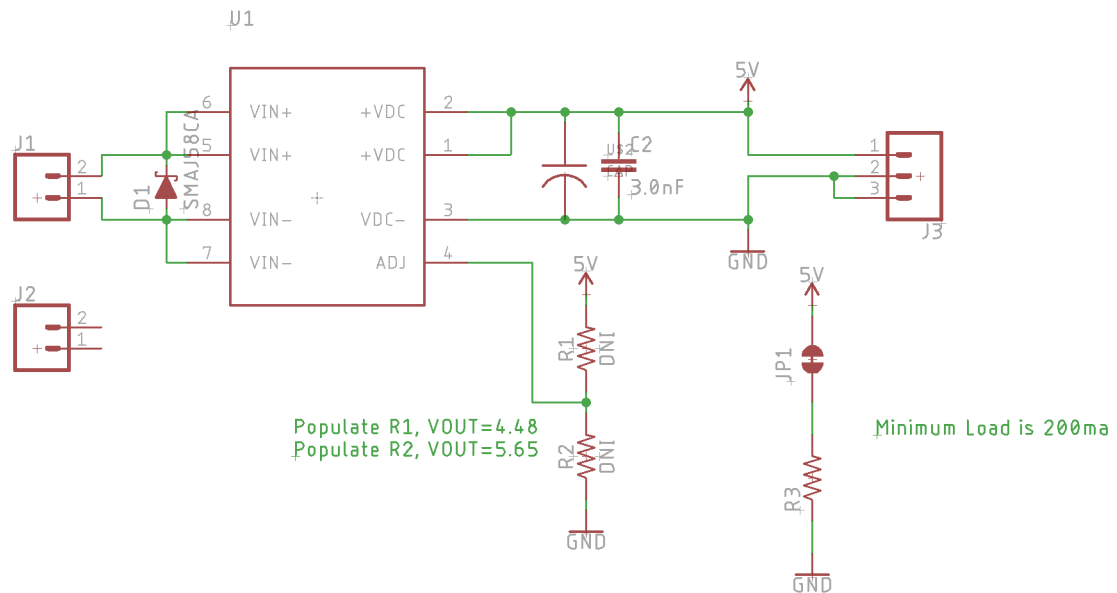
The PoE (Power over Ethernet) adapter for the Ethernet FeatherWing plugs into the Ethernet side of the FeatherWing.

It uses the Silvertel [Ag9905M](#) and can supply 1800 mA to your device and any Feather accessories. It is a Class 0 IEEE 802.3af device. It supports Alternative A (power on data pair) and Alternative B (power on spare pair).

Parameter	Minimum	Typical	Maximum	Units
Input Supply Voltage	36	48	57	Volts
Operating Temperature ¹	-40	25	70	°C
Output Voltage		5		V
Output Current			1800	mA

¹Operating temperature of the Ag9905M module only. The operating temperature range of Particle Gen 3 devices is smaller, -20 to +60°C.

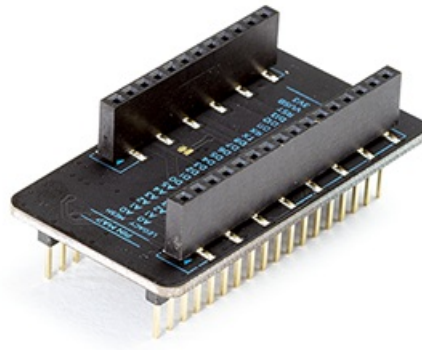
Note that because the Ethernet FeatherWing with PoE supplies 5V to the device by the VUSB pin, you should not power it by both PoE and the USB serial port at the same time.



The recommended cable for PoE is Cat 6 twisted pair with a maximum length of 100 meters (328 feet).

The Eagle CAD design for the PoE adapter is open source. All the files are available on our [GitHub repository](#).

Classic adapter



The Particle Classic Adapter lets you use your Electron Asset Tracker, Relay Shield, or other third-party shields with our third generation developer kits. Plug your Argon or Boron into this adapter and continue development as usual!

The top of the adapter has female connectors for you to plug in the Particle Gen 3 device. The bottom has male header pins in the Particle legacy form-factor for devices such as the Electron, Photon, or the Core.

The legacy side (male pins on the bottom) are designed to fit an Electron. So when plugging into a Photon socket please make sure the B/C pins side are hanging off the end, not plugged into anything.

You cannot supply more than 3.3V to any pin on the Argon or Boron (except VUSB and the USB connector).

Note: This product is a one-way adapter. That is, this product does not enable use of Adafruit FeatherWings with the Photon or Electron.

The design is open source and the design files are available [here](#).

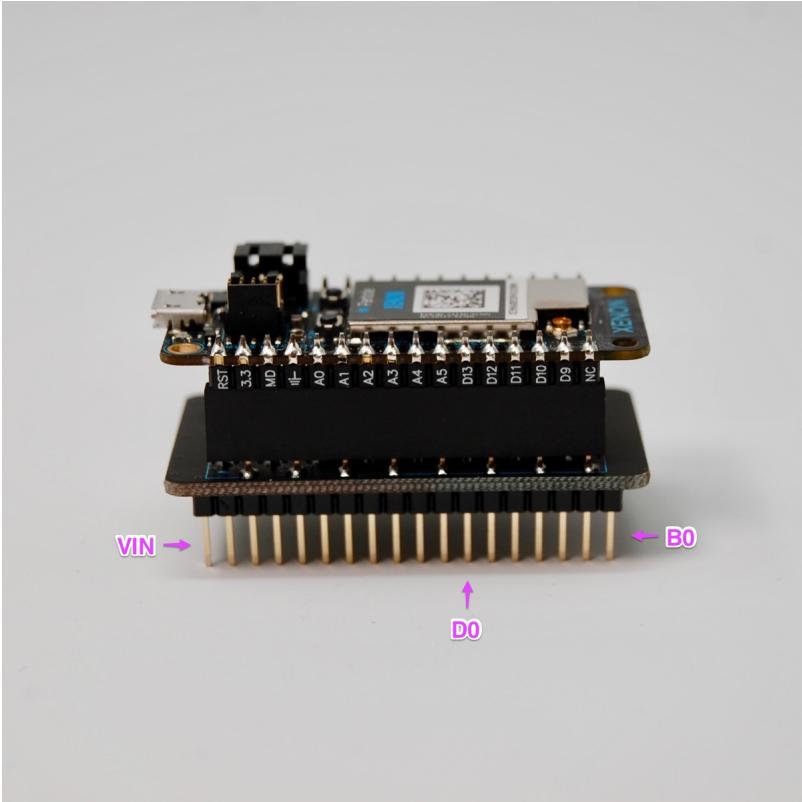
► Additional details and diagrams

Pin orientation

The long side (16-pin header on the top) has the pins arranged like this, left to right as in the picture. This only shows how the pins are oriented, not the connections!

Top (Gen 3)	Bottom (Gen 2)
RST	VIN
3V3	GND
MD	TX
GND	RX
A0	WKP
A1	DAC
A2	A5
A3	A4
A4	A3
A5	A2
D13 (SCK)	A1

D12 (MISO)	A0
D11 (MOSI)	B5
D10 (RX)	B4
D9 (TX)	B3
NC	B2
	B1
	B0

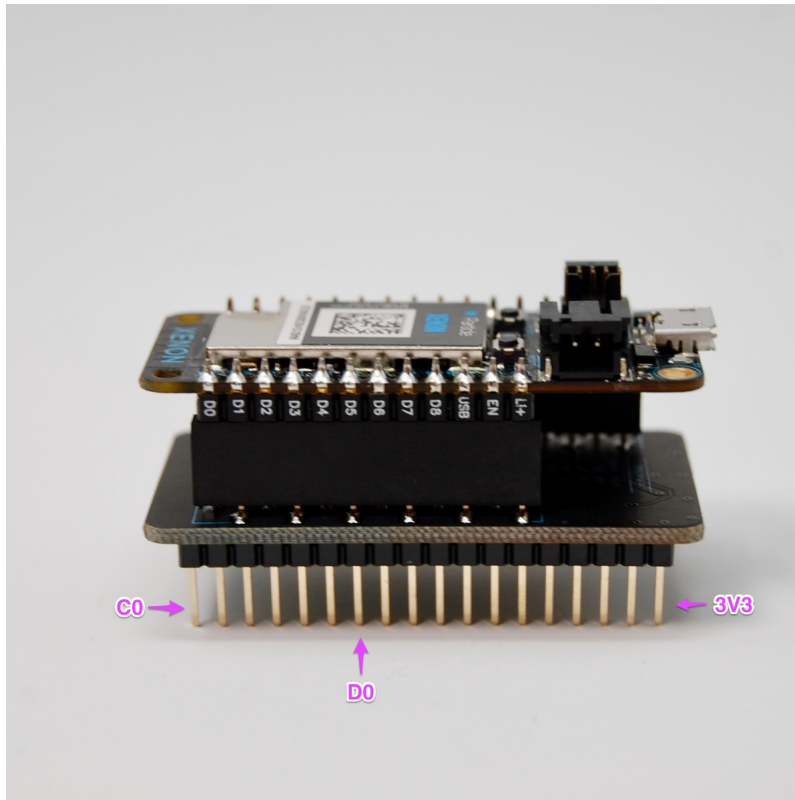


The short side (12-pin header on the top) has the pins arranged like this, left to right as in the picture.

Top (Gen 3)	Bottom (Gen 2)
D0	C0
D1	C1
D2	C2
D3	C3
D4	C4
D5	C5
D6	D0
D7	D1
D8	D2
USB	D3
EN	D4
LI+	D5
	D6
	D7
	GND
	VBAT

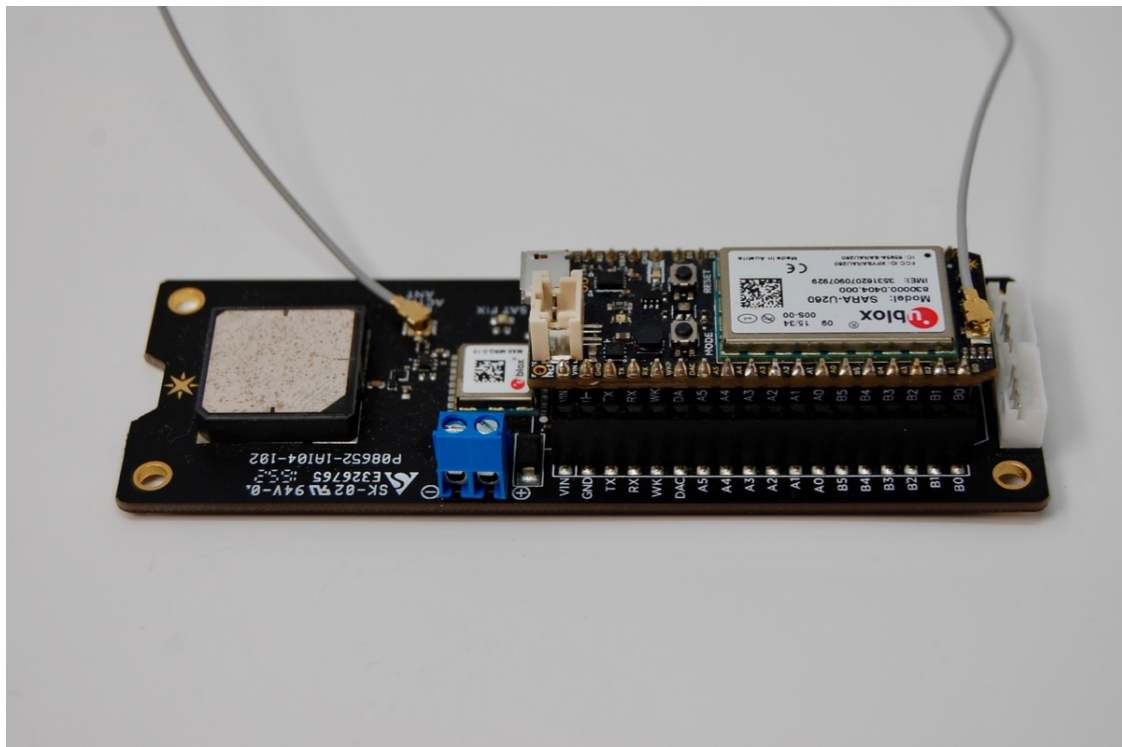
RST

3V3

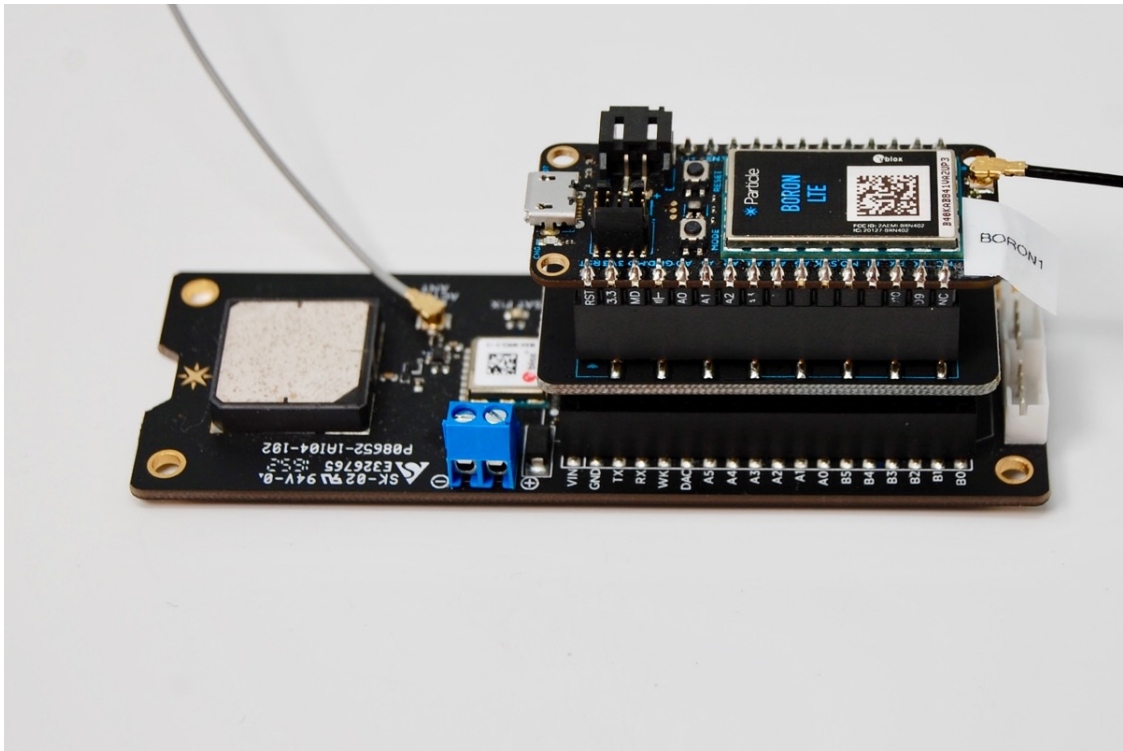


Substituting for an Electron

Take, for example, the Electron AssetTracker V2 with an Electron:



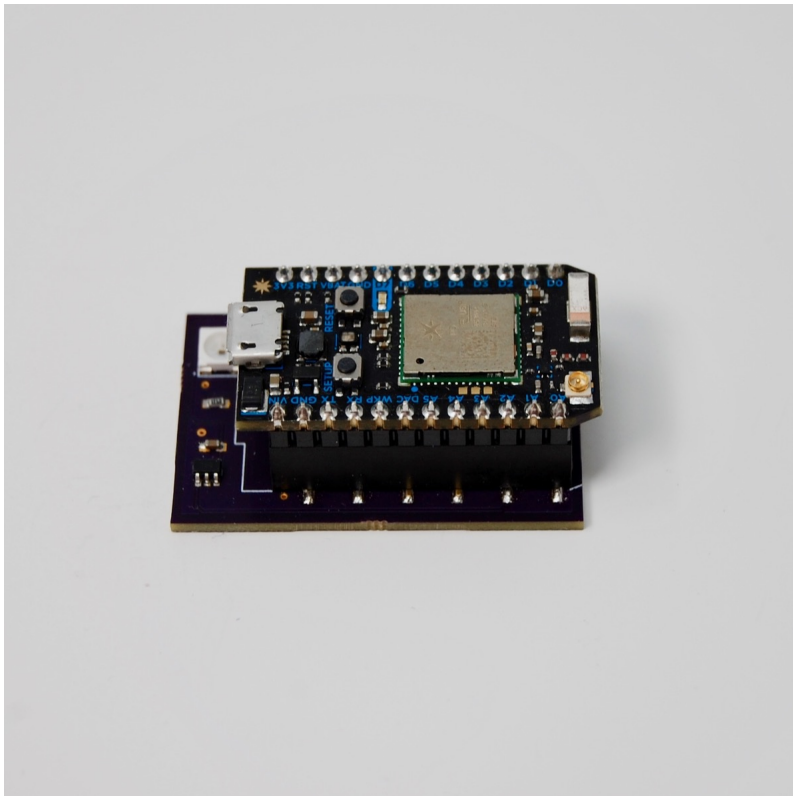
To substitute a Boron, you'd use the classic adapter like this:



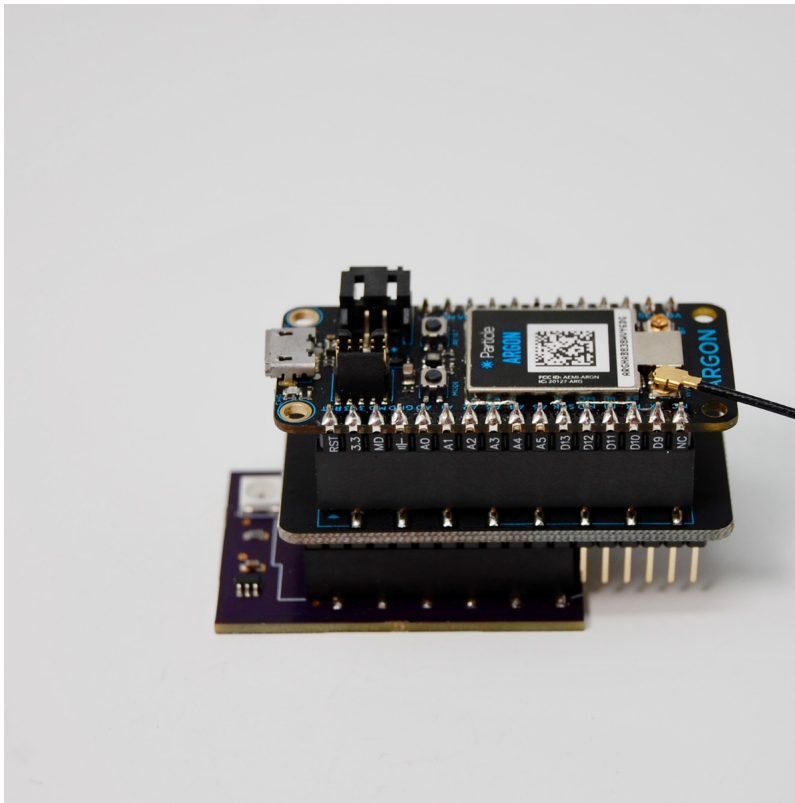
Note that you will not be able to use the LIS3DH accelerometer because it connects by SPI and the SPI pins are not mapped in a usable way with the classic adapter! You will also need to connect pin D6 to GND, which will cause the GPS to always be powered on. The reason is that if you turn the GPS on in software, the rush of current into it causes the nRF52 MCU to brown out and reset. If you turn them both on at the same time, both will successful start up.

Substituting for an Photon

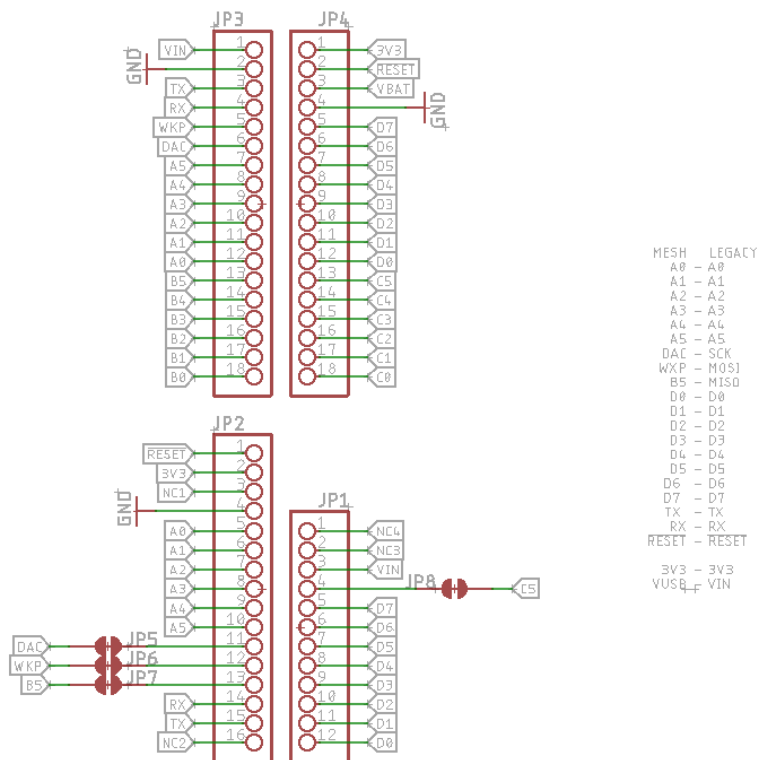
This is a small sensor that typically uses a Photon:



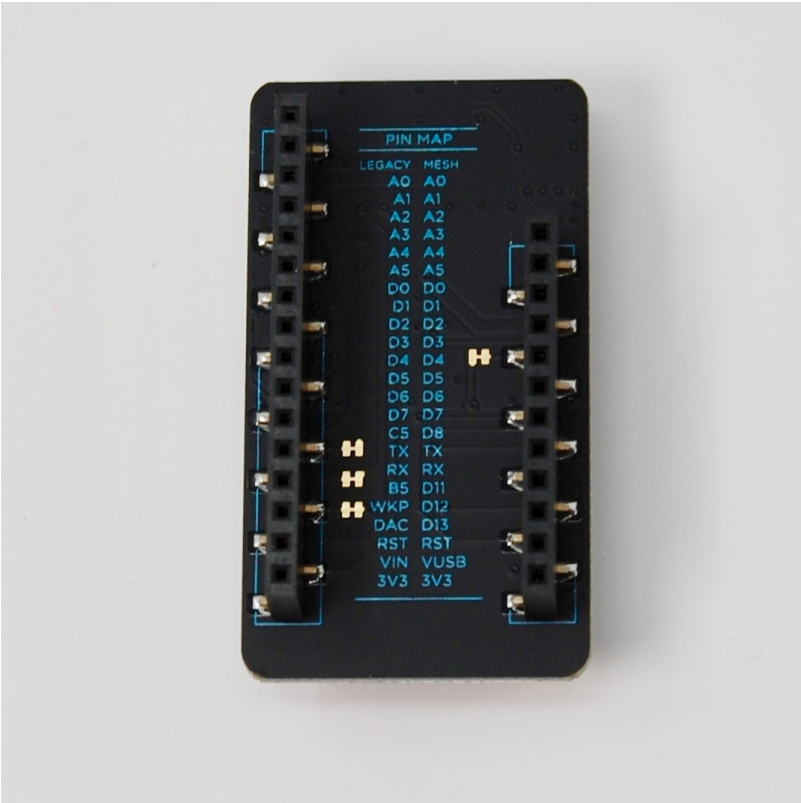
When using the classic adapter, some pins will hang off the edge. Makes sure the B and C pins hang over, not the power pins!



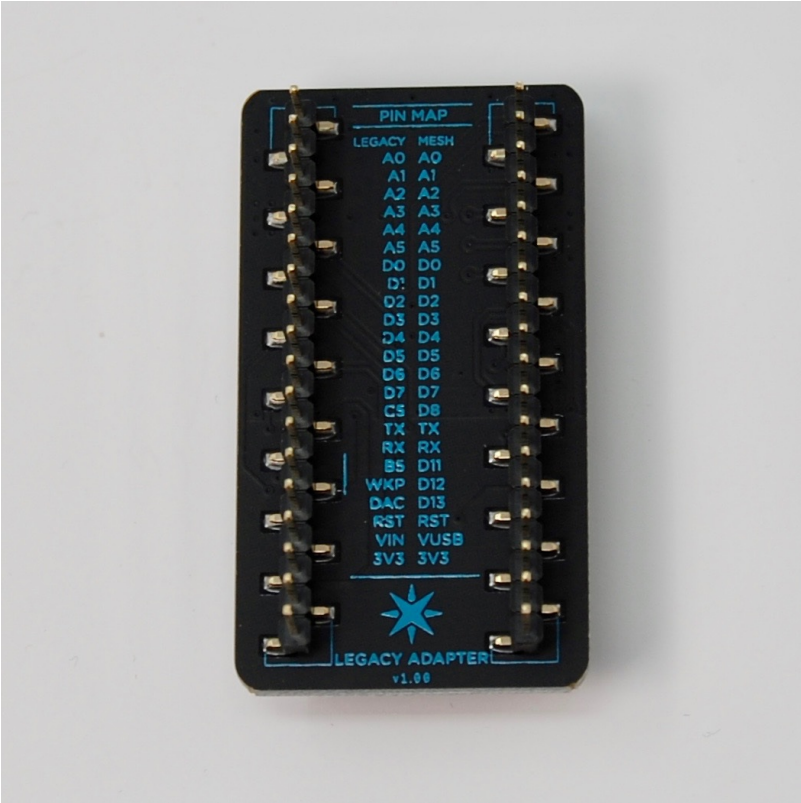
Schematic



Top



Bottom



PIN MAP

Legacy Device	Gen 3 Device
A0	A0
A1	A1

A2	A2
A3	A3
A4	A4
A5	A5
D0	D0 (SDA)
D1	D1 (SCL)
D2	D2
D3	D3
D4	D4
D5	D5
D6	D6
D7	D7
C5 ^[1]	D8
TX	TX
RX	RX
B5 ^[1]	D11 (MISO)
WKP ^[1]	D12 (MOSI)
DAC ^[1]	D13 (SCK)
RST	RST
VIN	VUSB
3V3	3V3

Notes:

^[1] These pins are connected via jumper pads on the top. You can cut these jumper traces and rewire them to a different pin on the Gen 3 device if the need arises.

^[2] Pins C0 to C4 and B0 to B4 on the Particle legacy device headers are not connected to anything.

^[3] Pins MD, Li+, and EN on the Particle Gen 3 device are not connected to anything.

I2C

I2C (Wire) can be used on D0 and D1. However, pins on the Gen 3 devices are not 5V tolerant, so make sure you do not have pull-ups to 5V.

SPI

It's not possible to use SPI with the classic adapter. The Gen 3 SPI pins (MISO, MOSI, SCK) are not connected to pins that support SPI on the Electron or Photon (B5, WKP, DAC). The Gen 3 SPI pins overlap the Electron/Photon SPI pins, however they're different!

SPI	Classic	Gen 3
SCK	D4	D2
MOSI	D2	D3
MISO	D3	D4

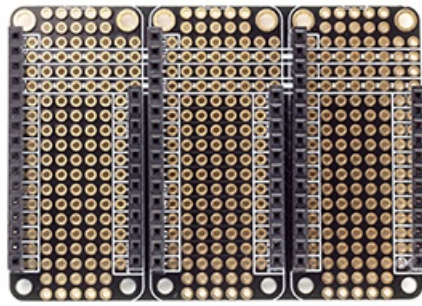
OTHER FEATURES

The Gen 3 devices do not support DAC or CAN, so those classic devices using those features won't work with the classic adapter.

Debugger

The [Particle Debugger](#) is now on its own page.

FeatherWing Tripler



Perfect for any prototyping project, the Tripler provides space to add-on up to two FeatherWing accessories without any stacking header needs. Compatible Gen 3 Particle developer kits.

The Tripler has three breakouts for each pin on a Feather, as well as plain grid proto holes. In addition, three sets of pins are cross connected with a full strip of connected pads for GND and 3.3V.